

Supplementary Information

scDesignGuard: source-bound design gating for multisample single-cell differential analysis

Supplementary Methods 1 | Framework, states, stage transitions and reason priority

Contracts bind source, donor, condition, target, support and estimand evidence. Gates are sequential. Terminal-state precedence is BLOCK, NON_EVALUABLE, ABSTAIN and PROCEED; activated reasons within a state are ordered lexicographically by frozen reason code. Unperformed checks remain NOT_ASSESSED or NOT_APPLICABLE.

Supplementary Methods 2 | Evidence tiers and authorization thresholds

Tier A, Tier B, Tier C and Tier D delimit confirmatory authorization, supportive use, structural use and controlled invalidity. Donor and target-cell cut-offs are conservative project rules, not universal sample-size recommendations.

Supplementary Methods 3 | Final 15-family overlap-excluded benchmark

Family identity was screened before endpoint estimation. Frozen source-bound human consensus labels and evaluator states are listed in Supplementary Table 1. Selection did not use evaluator state, downstream accessibility, biological-effect estimates or result appearance.

Supplementary Methods 4 | Four-case end-to-end evaluation

The four cases were processed in frozen order through source scope, structural design, target support, count aggregation, model validity and final authorization. Independent implementations reconciled counts and design-matrix quantities. No differential effect was estimated.

Supplementary Methods 5 | Controlled invalidities

Each challenge changed one design property and retained the remaining contract. Expected state and primary reason were fixed before evaluation.

Supplementary Methods 6 | Repository transport and software reproducibility

Transport audits used official identity and metadata endpoints without expression or outcomes. Software identity was bound to wheel, OCI image, schema, reason registry, SBOM, environment and test receipts.

Supplementary Methods 7 | Beta-cell application and sensitivity analyses

Author C0–C7 labels were retained. Donor-cluster pseudobulks supported composition, edgeR and limma-voom sensitivity analyses. Hallmark and Reactome CAMERA tests retained the full eligible multiple-testing universe; display subsets followed the frozen minimum-global-FDR ranking rule described in Methods.

Supplementary Table 1 | Final 15-family membership

record_id	repository	exact_source_version	frozen_human_reference_class	scdesignguard_primary_reason	scdesignguard_state	state_concordance_to_frozen_reference	endpoint_membership
DN12-GEO-01	GEO	GSE115978; source SHA-256=3735a895e33b6d4b13321e358d5bc0dbdec5eb61d9b6184	VALID_DESIGN	OUTPUT.SAFE_TO_TARGET_SUPPORT_GATE	PROCEED	YES	VALID_CLEARANCE;EXACT_STATE_CONCORDANCE

		a0343874a49791395					
DN12-GEO-03	GEO	GSE98638; source SHA-256=cca3a3e33fb5a9017609acac41d73b0e2620397f56130fe5e74af29b0e9fc698	VALID_DESIGN	OUTPUT.SAFE_TO_TARGET_SUPPORT_GATE	PROCEED	YES	VALID_CLEARANCE;EXACT_STATE_CONCORDANCE
DN12-GEO-04	GEO	GSE99254; source SHA-256=44ee6242f6e1f3b90215d4c69d7cd4ae76b52264e73a4f2325d8005d600d17eb	VALID_DESIGN	OUTPUT.SAFE_TO_TARGET_SUPPORT_GATE	PROCEED	YES	VALID_CLEARANCE;EXACT_STATE_CONCORDANCE
DN12-GEO-05	GEO	GSE81730; source SHA-256=6c8d1f937f495f5cd1ae95667e3c6307964bef65a12b55a1bfe23199ca960186	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNIT_S.BELOW_FROZEN_THRESHOLD	NON_EVALUABLE	YES	UNRESOLVED_CONTAINMENT;EXACT_STATE_CONCORDANCE
DN12-GEO-06	GEO	GSE72056; source SHA-256=a358716a7a6b88b8c3074fe628f5e23836e0f11b2a323668d284118d0302f280	STRUCTURALLY_UNRESOLVED	ESTIMAND.UNRESOLVED	NON_EVALUABLE	YES	UNRESOLVED_CONTAINMENT;EXACT_STATE_CONCORDANCE
DN12-GEO-07	GEO	GSE57872; source SHA-256=35157d95b27a56c7b5e8d4bc33ee3e9bc4a34dc4466c71ace6cfc8ff90332f81	STRUCTURALLY_UNRESOLVED	SOURCE_SCOPE.MIXED_OBJECTS_AND_ESTIMAND_UNRESOLVED	NON_EVALUABLE	YES	UNRESOLVED_CONTAINMENT;EXACT_STATE_CONCORDANCE
DN12-GEO-08	GEO	GSE85241; source SHA-256=b7f6c310a48d5f69b730da1fda750100d08546e1ed5a99faec85a8bd66b0a6cc	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNIT_S.BELOW_FROZEN_THRESHOLD	NON_EVALUABLE	YES	UNRESOLVED_CONTAINMENT;EXACT_STATE_CONCORDANCE
DN12-GEO-09	GEO	GSE78711; source SHA-256=4757b3f5eb21c62c30ed4ef4d71db525419b3791268dcf853b90e786f0a1880d	HARD_INVALID_DESIGN	SOURCE_SCOPE.NOT_SINGLE_CELL	BLOCK	YES	HARD_INVALID_DETECTION;UNSAFE_CONTINUATION;EXACT_STATE_CONCORDANCE
DN12-GEO-10	GEO	GSE79136; source SHA-256=31686ddaca23d1fa5d8c6eb153d0d317b7564724405a6a8212b9217bcd171de5	HARD_INVALID_DESIGN	DESIGN.NO_DONOR_LEVEL_BIOLOGICAL_ESTIMAND	BLOCK	YES	HARD_INVALID_DETECTION;UNSAFE_CONTINUATION;EXACT_STATE_CONCORDANCE
DN12-GEO-11	GEO	GSE85575; source SHA-256=af7440179e7271dda4d6b12a4555c9352f8a7abc270a71bc743e9a8ebce1c88b	HARD_INVALID_DESIGN	DESIGN.CELL_LINE_TECHNICAL_BENCHMARK	BLOCK	YES	HARD_INVALID_DETECTION;UNSAFE_CONTINUATION;EXACT_STATE_CONCORDANCE
DN12-GEO-12	GEO	GSE75748; source SHA-256=6de92677aa582fbfcf955d65ea2abe958a4f13f27848aec5fe63a712f4b2aab	HARD_INVALID_DESIGN	DESIGN.NO_INDEPENDENT_HUMAN_DONOR_REPLICATION	BLOCK	YES	HARD_INVALID_DETECTION;UNSAFE_CONTINUATION;EXACT_STATE_CONCORDANCE

		6					
DN12-AE-01	BioStudies/ ArrayExpress	E-MTAB-5905; source SHA-256=47151 81fcf871b8467c 98578f8715e9f8e 3a5ba517f1a1d3 bb284eaba72884 9	VALID_DESIGN	OUTPUT.SAFE_ TO_TARGET_SU PPORT_GATE	PROCEED	YES	VALID_CLEARA NCE;EXACT_ST ATE_CONCORD ANCE
DN12-AE-02	BioStudies/ ArrayExpress	E-MTAB-6653; source SHA-256=535b6 962712fb4ec0c9 d0e5286816796e 08bc6d9128e1e3 2196b2492cf5e6 a86	STRUCTURALLY _UNRESOLVED	SUPPORT.INDE PENDENT_UNIT S.BELOW_FROZ EN_THRESHOLD	NON_EVALUAB LE	YES	UNRESOLVED_ CONTAINMENT; EXACT_STATE_ CONCORDANCE
DN12-AE-03	BioStudies/ ArrayExpress	E-MTAB-10296; source SHA-256=5907e 283cf857b3e0c8 2b671fced8bb97 d4507dda77ca6b 7eda6186fc845d 05	HARD_INVALID_ DESIGN	DESIGN.CELL_L INE_METHOD_D EMONSTRATIO N	BLOCK	YES	HARD_INVALID_ DETECTION;UN SAFE_CONTINU ATION;EXACT_S TATE_CONCOR DANCE
DN12-AE-04	BioStudies/ ArrayExpress	E-MTAB-5480; source SHA-256=12fe98 0c61b63fd33420 933634c1149e87 c2e2691cbb8fe6 58243a920a6dab f0	HARD_INVALID_ DESIGN	DESIGN.TECHNI CAL_METHOD_ COMPARISON	BLOCK	YES	HARD_INVALID_ DETECTION;UN SAFE_CONTINU ATION;EXACT_S TATE_CONCOR DANCE

Supplementary Table 2 | Final four-case end-to-end ledger

display_id	accession_or_collection_id	request_started_utc	contrast	target	donor_count	pair_count	target_cell_count	pseudo_bulk_rows	design_rank	estimability	residual_df	terminal_result
GSE144735	GSE144735	2026-08-09T08:45:24Z	paired condition contrast	epithelial cells	6	6	3356	12	7/7	YES	5	PASS_FINAL_AUTHORIZATION_GATE_NO_DEOR_EFFECT_EXECUTED
GSE123813	GSE123813	2026-08-09T08:45:25Z	paired condition contrast	T cells	11	11	33106	22	12/12	YES	10	PASS_FINAL_AUTHORIZATION_GATE_NO_DEOR_EFFECT_EXECUTED
HCA-E2E-03	135f7f5c-4a85-4bcf-9f7c-4f035ff1e428	2026-08-09T08:45:31Z	source-defined comparison	NOT_ASSESSED	NOT_ASSESSED	NOT_ASSESSED	NOT_ASSESSED	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	EARLY_STOP_STRUCTUREL_DESIGN_SOURCE_CONTRACT_INCCOMPLETE
GSE81076	GSE81076	2026-08-09T08:45:35Z	independent groups	source-defined	5	0	NOT_ASSESSED	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	EARLY_STOP_STRUCTUREL_DESIGN_Below_FROZEN_GROUP_THRESHOLD

Supplementary Table 3 | Frozen state and reason hierarchy

state_precedence	within_state_priority	reason_code	terminal_state	definition	primary_reason_rule
1	1	COUNT.SOURCE.INVALID	BLOCK	The frozen count representation is contract-invalid.	state precedence, then lexicographic reason code
1	2	DESIGN.CONDITION.CONFLICT	BLOCK	A donor maps to conflicting frozen conditions.	state precedence, then lexicographic reason code
1	3	DESIGN.CONFOUNDING.COMPLETE	BLOCK	The estimand is inseparable from complete technical confounding.	state precedence, then lexicographic reason code
1	4	DESIGN.DONOR.INVALID	BLOCK	Donor is not the inference unit or repeated samples are treated as donors.	state precedence, then lexicographic reason code
1	10	GOV.CLAIM.UNMAPPED	BLOCK	A proposed release claim is not mapped to a passed endpoint and Gate.	state precedence, then lexicographic reason code
1	11	GOV.IDENTITY.UNBOUND	BLOCK	Immutable identity/version/hash is not exactly bound.	state precedence, then lexicographic reason code
1	12	GOV.LICENSE.UNRESOLVED	BLOCK	Use rights or redistribution terms are	state precedence, then lexicographic reason

				unresolved.	code
1	13	GOV.ROLE.LEAKAGE	BLOCK	Outcome access or role assignment occurred before freeze.	state precedence, then lexicographic reason code
2	5	EVIDENCE.ESTIMABILITY.INSUFFICIENT	NON_EVALUABLE	The frozen contrast or endpoint is not estimable.	state precedence, then lexicographic reason code
2	6	EVIDENCE.INDEPENDENCE.INSUFFICIENT	NON_EVALUABLE	Required independent evidence is unavailable or overlaps.	state precedence, then lexicographic reason code
2	7	EVIDENCE.MODEL.NO_NESTIMABLE	NON_EVALUABLE	The frozen statistical model cannot be estimated without amendment.	state precedence, then lexicographic reason code
2	8	EVIDENCE.REFERENCE.MISSING	NON_EVALUABLE	A required reference measurement is absent.	state precedence, then lexicographic reason code
2	9	EVIDENCE.TARGET_SUPPORT.INSUFFICIENT	NON_EVALUABLE	Prespecified donor-target support is insufficient.	state precedence, then lexicographic reason code
3	15	ROBUSTNESS.ANNOTATION.SENSITIVE	ABSTAIN	Conclusion is sensitive to prespecified annotation perturbation.	state precedence, then lexicographic reason code
3	16	ROBUSTNESS.CONFOUNDING.SENSITIVE	ABSTAIN	Conclusion is sensitive to partial confounding adjustment.	state precedence, then lexicographic reason code
3	17	ROBUSTNESS.DIRECTION.UNSTABLE	ABSTAIN	Direction is unstable under the frozen robustness profile.	state precedence, then lexicographic reason code
3	18	ROBUSTNESS.FILTERING.FRAGILE	ABSTAIN	Eligibility/filtering is unstable under prespecified perturbation.	state precedence, then lexicographic reason code
3	19	ROBUSTNESS.LEAVE-ONE-DONOR-OUT.ROBUSTNESS.FAILS	ABSTAIN	Prespecified leave-one-donor-out robustness fails.	state precedence, then lexicographic reason code
3	20	UNCERTAINTY.CLUSTERING.UNRESOLVED	ABSTAIN	Study/donor clustered uncertainty cannot be resolved.	state precedence, then lexicographic reason code
3	21	UNCERTAINTY.COVERAGE.BELOW_THRESHOLD	ABSTAIN	Prespecified uncertainty coverage rule fails.	state precedence, then lexicographic reason code
4	14	OUTPUT.SAFE_TO_PROCEED	PROCEED	All higher-precedence reasons are absent.	state precedence, then lexicographic reason code

Supplementary Table 4 | Controlled invalidities

challenge_id	challenge	terminal_state	expected_primary_reason	primary_reason_code	safe_nonproceed	exact_primary_reason_match
KIC-01	DONOR_DELETION	NON_EVALUABLE	EVIDENCE.TARGET_SUPPORT.INSUFFICIENT	EVIDENCE.TARGET_SUPPORT.INSUFFICIENT	YES	YES
KIC-02	CELL_LEVEL_PSEUDOREPLICATION	BLOCK	DESIGN.DONOR.INVALID	DESIGN.DONOR.INVALID	YES	YES
KIC-03	BATCH_CONDITION_CONFOUNDING	BLOCK	DESIGN.CONFOUNDING.COMPLETE	DESIGN.CONFOUNDING.COMPLETE	YES	YES
KIC-04	DUPLICATE_DONOR	BLOCK	DESIGN.DONOR.INVALID	DESIGN.DONOR.INVALID	YES	YES
KIC-05	PAIR_DESTRUCTION	NON_EVALUABLE	EVIDENCE.MODEL.NONESTIMABLE	EVIDENCE.MODEL.NONESTIMABLE	YES	YES
KIC-06	WRONG_TARGET	NON_EVALUABLE	EVIDENCE.REFERENCE.MISSING	EVIDENCE.REFERENCE.MISSING	YES	YES
KIC-07	WRONG_VERSION	BLOCK	GOV.IDENTITY.UNBOUND	GOV.IDENTITY.UNBOUND	YES	YES
KIC-08	MIXED_STRATA	BLOCK	DESIGN.CONDITION	DESIGN.CONDITION	YES	YES

			N.CONFLICT	N.CONFLICT		
KIC-09	LOW_SUPPORT	NON_EVALUABLE	EVIDENCE.TARGET_SUPPORT.INSUFFICIENT	EVIDENCE.TARGET_SUPPORT.INSUFFICIENT	YES	YES

Supplementary Table 5 | Repository transport audit

transport_audit_id	study_family_id	repository_platform	publication_year	time_split	initial_tier	terminal_state	reason_codes	expression_accessed	outcome_accessed	effect_claim
HOLD-001	SF-CXG-3116d060-0a8e-4767-99bbe866badea1ed	BIOSTUDIES_ARRAYEXPRESSES	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-002	SF-CXG-b1a879f6-5638-48d3-8f64-f6592c1b1561	BIOSTUDIES_ARRAYEXPRESSES	2022	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-003	SF-CXG-e2c257e7-6f79-487c-b81c-39451cd4ab3c	BIOSTUDIES_ARRAYEXPRESSES	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-004	SF-CXG-2d40e6a7-f2fd-49ba-9db9-6b97e4c6dad5	BIOSTUDIES_ARRAYEXPRESSES	2024	2024_OR_LATER	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-005	SF-CXG-73f82ac8-15cc-4fcd-87f8-5683723fce7f	BIOSTUDIES_ARRAYEXPRESSES	2024	2024_OR_LATER	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-006	SF-CXG-02b01703-bf1b-48deb99a-23bef8cccc81	SINGLE_CELL_PORTAL	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-007	SF-CXG-fe0e718d-2ee9-42cc-894b-0b490f437dfd	SINGLE_CELL_PORTAL	2024	2024_OR_LATER	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAPPED;DONOR_MAPPING_NOT_EVALUATED;SUPPORT_NOT_COMPUTED	NO	NO	PROHIBITED
HOLD-008	SF-CXG-33d19	SINGLE_CELL_PORTAL	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAP	NO	NO	PROHIBIT

	f34-87f5-455b-8ca5-9023a2e5453d	AL			AL	LUABLE	PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED			ED
HOLD-009	SF-CXG-a3ffde6c-7ad2-498a-903cd58e732f7470	SINGLE_CELL_PORTAL	2022	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAP PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED	NO	NO	PROHIBITED
HOLD-010	SF-CXG-8e880741-bf9a-4c8e-9227-934204631d2a	NCBI_GEO	2021	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAP PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED	NO	NO	PROHIBITED
HOLD-011	SF-CXG-74e10dc4-cbb2-4605-a189-8a1cd8e44d8c	NCBI_GEO	2023	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAP PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED	NO	NO	PROHIBITED
HOLD-012	SF-CXG-bfd55632-15e2-4dde-a1b9-02ade511fc4d	NCBI_GEO	2025	2024_OR_LATER	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAP PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED	NO	NO	PROHIBITED
HOLD-013	SF-CXG-60358420-6055-411d-ba4fe8ac80682a2e	NCBI_GEO	2021	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAP PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED	NO	NO	PROHIBITED
HOLD-014	SF-CXG-ba84c7ba-8d8c-4720-a76e-3ee37dc89f0b	NCBI_GEO	2024	2024_OR_LATER	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAP PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED	NO	NO	PROHIBITED
HOLD-015	SF-CXG-e5f58829-1a66-40b5-a624-9046778e74f5	HUMAN_CELL_ATLAS	2021	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAP PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED	NO	NO	PROHIBITED
HOLD-016	SF-CXG-b52eb423-5d0d-4645-b217-e1c6d38b2e72	HUMAN_CELL_ATLAS	2021	PRE_2024	TIER_C_STRUCTUREAL	NON_EVALUABLE	ONTOLOGY_UNMAP PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED	NO	NO	PROHIBITED
HOLD-017	SF-	HUMAN_C	2022	PRE_2024	TIER_C_S	NON_EVA	ONTOLOG	NO	NO	PROHIBIT

	CXG-120e8 6b4-1195- 48c5-845b- b98054105 eec	ELL_ATLA S			TRUCTUR AL	LUABLE	Y_UNMAP PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED			ED
HOLD-018	SF- CXG-3f503 14f-bdc9- 40c6-8e4a- b0901ebfbc 4c	HUMAN_C ELL_ATLA S	2023	PRE_2024	TIER_C_S TRUCTUR AL	NON_EVA LUABLE	ONTOLOG Y_UNMAP PED;DON OR_MAPPI NG_NOT_ EVALUATE D;SUPPOR T_NOT_C OMPUTED	NO	NO	PROHIBIT ED

Supplementary Table 6 | Software identity and reproducibility

artifact_type	archive_path	bytes	sha256	evaluated_so urce_commit	evaluated_so urce_tree	identity_stat us	public_relea se_status	interpretive_ boundary
wheel	software_revie wer_archive/ artifacts/ scdesignguard _nm03-0.1.0- py3-none- any.whl	15767	59b74e2e6031 5094fb5e9d70 224eb7724d88 39d75e2e3f84 57042f2df0ec9 986	6d7d87b45c64 c7f6f62ba818e b69b33c147d6 6e5	f2be0de9ac1c0 d6cb55274d20 9254cab5b0f3f 68	ARCHIVAL_ID ENTITY_BOU ND	NOT_PUBLICL Y_RELEASED _REVIEWER_ ARCHIVE_ON LY	ARCHIVAL_ID ENTITY_MANI FEST_NOT_R ETROSPECTI VE_PROSPEC TIVE_FREEZE
oci_image	software_revie wer_archive/ artifacts/ scdesignguard _nm03_0.1.0.o ci.tar	50994176	ad675089b934 e6d96c1bfd83 b9deb4fc6c9ac 8dbda345ce9e 08a9d7d9b21c 85d	6d7d87b45c64 c7f6f62ba818e b69b33c147d6 6e5	f2be0de9ac1c0 d6cb55274d20 9254cab5b0f3f 68	ARCHIVAL_ID ENTITY_BOU ND	NOT_PUBLICL Y_RELEASED _REVIEWER_ ARCHIVE_ON LY	ARCHIVAL_ID ENTITY_MANI FEST_NOT_R ETROSPECTI VE_PROSPEC TIVE_FREEZE
json_schema	software_revie wer_archive/ artifacts/ design- contract.schem a.v1.0.0.json	4259	8755803b1a02 24f5d41ab283 a9ea3fe60722 8f40f4fec8d51c 514d9b682946 16	6d7d87b45c64 c7f6f62ba818e b69b33c147d6 6e5	f2be0de9ac1c0 d6cb55274d20 9254cab5b0f3f 68	ARCHIVAL_ID ENTITY_BOU ND	NOT_PUBLICL Y_RELEASED _REVIEWER_ ARCHIVE_ON LY	ARCHIVAL_ID ENTITY_MANI FEST_NOT_R ETROSPECTI VE_PROSPEC TIVE_FREEZE
reason_code_r egistry	software_revie wer_archive/ REASON_CO DE_MANIFES T.csv	1987	7317bfe7c11bc d960ddd60345 fd5ef71928b5c 3ed2afd23133f 62b6f2149b9f3	6d7d87b45c64 c7f6f62ba818e b69b33c147d6 6e5	f2be0de9ac1c0 d6cb55274d20 9254cab5b0f3f 68	ARCHIVAL_ID ENTITY_BOU ND	NOT_PUBLICL Y_RELEASED _REVIEWER_ ARCHIVE_ON LY	ARCHIVAL_ID ENTITY_MANI FEST_NOT_R ETROSPECTI VE_PROSPEC TIVE_FREEZE
dependency_d efinition	software_revie wer_archive/ pyproject.toml	649	56d8902896a2 d26fe63baf2a8 2f7641d617ab 6d8bfa944124 4f166f007d15d 28	6d7d87b45c64 c7f6f62ba818e b69b33c147d6 6e5	f2be0de9ac1c0 d6cb55274d20 9254cab5b0f3f 68	ARCHIVAL_ID ENTITY_BOU ND	NOT_PUBLICL Y_RELEASED _REVIEWER_ ARCHIVE_ON LY	ARCHIVAL_ID ENTITY_MANI FEST_NOT_R ETROSPECTI VE_PROSPEC TIVE_FREEZE
container_defin ition	software_revie wer_archive/ Containerfile	543	afb6cdf276241 827ffc5bb8a7d 44a7345bb382 e43977cfcc75e cdd503f1743 8	6d7d87b45c64 c7f6f62ba818e b69b33c147d6 6e5	f2be0de9ac1c0 d6cb55274d20 9254cab5b0f3f 68	ARCHIVAL_ID ENTITY_BOU ND	NOT_PUBLICL Y_RELEASED _REVIEWER_ ARCHIVE_ON LY	ARCHIVAL_ID ENTITY_MANI FEST_NOT_R ETROSPECTI VE_PROSPEC TIVE_FREEZE
sbom	software_revie wer_archive/ artifacts/ SBOM.spdx.js on	9432	e02b3637b244 dba1002bf221 c479f289cb427 0189d39922cf 929adf02f52aa 68	6d7d87b45c64 c7f6f62ba818e b69b33c147d6 6e5	f2be0de9ac1c0 d6cb55274d20 9254cab5b0f3f 68	ARCHIVAL_ID ENTITY_BOU ND	NOT_PUBLICL Y_RELEASED _REVIEWER_ ARCHIVE_ON LY	ARCHIVAL_ID ENTITY_MANI FEST_NOT_R ETROSPECTI VE_PROSPEC TIVE_FREEZE
test_receipt	software_revie wer_archive/ independent_r eproduction_re ceipts/ VALIDATION_ REPORT.json	7007	0d67fa2be499 819b61757651 0b1565673c59 8c53c93aaa2 2efcf7f313e071 dcb	6d7d87b45c64 c7f6f62ba818e b69b33c147d6 6e5	f2be0de9ac1c0 d6cb55274d20 9254cab5b0f3f 68	ARCHIVAL_ID ENTITY_BOU ND	NOT_PUBLICL Y_RELEASED _REVIEWER_ ARCHIVE_ON LY	ARCHIVAL_ID ENTITY_MANI FEST_NOT_R ETROSPECTI VE_PROSPEC TIVE_FREEZE
execution_rece ipt	software_revie wer_archive/ 	924	719e7f2f7f1c5c 13ffa02bd8295	6d7d87b45c64 c7f6f62ba818e	f2be0de9ac1c0 d6cb55274d20	ARCHIVAL_ID ENTITY_BOU	NOT_PUBLICL Y_RELEASED	ARCHIVAL_ID ENTITY_MANI

	independent_reproduction_receipts/W6C_EXECUTION_RECEIPT.json		9a2ff829f7d2a39f40b36aea4d3eca1144579	b69b33c147d66e5	9254cab5b0f3f68	ND	_REVIEWER_ARCHIVE_ONLY	FEST_NOT_RETROSPECTIVE_PROSPECTIVE_FREEZE
security_receipt	software_reviewer_archive/independent_reproduction_receipts/SBOM_SECURITY_RECEIPT.json	1295	7877a925f17b906fd4ca9aa4b088d363940dbd70a649c5aebc27d38c6cafbcd9	6d7d87b45c64c7f6f62ba818eb69b33c147d66e5	f2be0de9ac1c0d6cb55274d209254cab5b0f3f68	ARCHIVAL_IDENTITY_BOUNDED	NOT_PUBLICLY_RELEASED_REVIEWER_ARCHIVE_ONLY	ARCHIVAL_IDENTITY_MANIFEST_NOT_RETROSPECTIVE_PROSPECTIVE_FREEZE

metric	numerator	denominator	detail
current_release_test_suites	3	3	PASS
expected_historical_compatibility_suites	2	2	preserved behavior
repository_unit_tests	1362	1362	PASS

Supplementary Table 7 | Beta-cell analysis, quality control and sensitivity

cluster	ND	PD	T2D	contrast	tested_genes	edgeR_within_FD_R_0_05	edgeR_global_FD_R_0_05	voom_global_FD_R_0_05	global_overlap	spearman_logFC	sign concordance
C0	17	14	16	T2D-ND	14259	79	71	72	53	0.9248869538485428	0.9120555438670314
C1	17	14	17	T2D-ND	14647	158	110	120	93	0.9393810206372684	0.9281764183791904
C2	17	14	17	T2D-ND	13918	103	82	90	58	0.943957471504516	0.9295875844230492
C3	17	14	16	T2D-ND	12549	60	58	63	45	0.9544616135062206	0.9291577018089092
C4	17	14	17	T2D-ND	11603	17	28	26	18	0.953515298017543	0.9309661294492804
C5	17	14	17	T2D-ND	12367	26	36	34	24	0.9320549161906678	0.920109970081669
C6	17	14	17	T2D-ND	12805	19	30	31	17	0.9268520703689374	0.9169074580242093
C7	13	8	10	T2D-ND	8286	0	3	8	3	0.9483913150964532	0.9240888245232924

Supplementary Table 8 | Conservative capability comparison

approach	documented_primary_role	source_version_gate	donor_design	ontology_relation	target_support	estimability_authorization	effect_estimation	evidence
scDesignGuard	Pre-inference source/design authorization	Required	Required	Required	Required	Required	Delegates estimation to downstream engines	This study
Donor-aware pseudobulk practice	Biological-unit-aware aggregation and inference	Not a required or documented core function	Core design requirement	Not a required or documented core function	Study-specific	Model-dependent	Performs effect estimation through a statistical	Refs. 8–10

							engine	
edgeR	Count-based differential analysis	Not a required or documented core function	Supplied through the design matrix	Not a required or documented core function	Not a required or documented core function	Model and contrast diagnostics	Performs effect estimation	Ref. 20
limma-voom	Precision-weighted linear modelling	Not a required or documented core function	Supplied through the design matrix	Not a required or documented core function	Not a required or documented core function	Model and contrast diagnostics	Performs effect estimation	Refs. 52,53
Cell Ontology / SSSOM	Semantic identity and mapping representation	Not a required or documented core function	Not a required or documented core function	Core semantic function	Not a required or documented core function	Not a required or documented core function	Does not estimate effects	Refs. 46–48

Entries describe documented primary roles and required core functions. They do not constitute a performance ranking or a claim that other approaches cannot be extended.

Supplementary Table 9 | Registered 24-family selection and overlap-exclusion ledger

This exact candidate ledger records the frozen 24-family registry, the nine prespecified development/calibration study-family overlap exclusions and the 15 families contributing to final primary performance denominators. Excluded records retain their frozen human labels and contribute zero to every primary numerator and denominator.

registered_order	record_id	repository	accession_or_collection_id	frozen_human_reference_class	frozen_human_primary_reason	final_overlap_status	final_primary_inclusion	exclusion_reason	primary_denominator_contribution
1	DN12-GEO-01	GEO	GSE115978	VALID_DESIGN	NONE_STRUCTURE_C LEAR	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
2	DN12-GEO-02	GEO	GSE131907	VALID_DESIGN	NONE_STRUCTURE_C LEAR	DEVELOPMENT_CORP US_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
3	DN12-GEO-03	GEO	GSE98638	VALID_DESIGN	NONE_STRUCTURE_C LEAR	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
4	DN12-GEO-04	GEO	GSE99254	VALID_DESIGN	NONE_STRUCTURE_C LEAR	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
5	DN12-GEO-05	GEO	GSE81730	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNITS.BELOW_FROZEN_THRESHOLD	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
6	DN12-GEO-06	GEO	GSE72056	STRUCTURALLY_UNRESOLVED	ESTIMAND.UNRESOLVED	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
7	DN12-GEO-07	GEO	GSE57872	STRUCTURALLY_UNRESOLVED	SOURCE_SCOPE.MIXED_OBJECTS_AND_ESTIMAND_UNRESOLVED	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
8	DN12-GEO-08	GEO	GSE85241	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNITS.BELOW_FROZEN_THRESHOLD	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
9	DN12-GEO-09	GEO	GSE78711	HARD_INVALID_DESIGN	SOURCE_SCOPE.NOT_SINGLE_CELL	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
10	DN12-GEO-10	GEO	GSE79136	HARD_INVALID_DESIGN	DESIGN.NO_DONOR_LEVEL_BIOLOGICAL_ESTIMAND	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
11	DN12-GEO-11	GEO	GSE85575	HARD_INVALID_DESIGN	DESIGN.CELL_LINE_TECHNICAL_BENCHMARK	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
12	DN12-GEO-12	GEO	GSE75748	HARD_INVALID_DESIGN	DESIGN.NO_INDEPENDENT_HUMAN_DONOR_REPLICATION	NO_EXACT_ID_MATCH _IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INC LUDED	1
13	DN12-CXG-01	CELLxGENE	07e12576-b41b-4350-adfc-059bfc4328ea	VALID_DESIGN	NONE_STRUCTURE_C LEAR	DEVELOPMENT_CORP US_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
14	DN12-CXG-02	CELLxGENE	7c7bd6c2-925b-4034-baab-620ef1b760e1	VALID_DESIGN	NONE_STRUCTURE_C LEAR	DEVELOPMENT_CORP US_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
15	DN12-CXG-03	CELLxGENE	1ca90a2d-2943-483d-	VALID_DESIGN	NONE_STRUCTURE_C	DEVELOPMENT_CORP	NO	PREDEFINED_DEVELOPMENT	0

registered_order	record_id	repository	accession_or_collection_id	frozen_human_reference_class	frozen_human_primary_reason	final_overlap_status	final_primary_inclusion	exclusion_reason	primary_denominator_contribution
			b678-b809bf464c30		LEAR	US_EXACT_ID_OVERLAP_FAIL_CLOSED		PMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	
16	DN12-CXG-04	CELLxGENE	c53573b2-eff4-4c5e-9ad0-b24d422dfd9b	VALID_DESIGN	NONE_STRUCTURE_CLEAR	DEVELOPMENT_CORPUS_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
17	DN12-CXG-05	CELLxGENE	84ce6837-548d-4a1f-919f-0bc0d9a3952f	VALID_DESIGN	NONE_STRUCTURE_CLEAR	DEVELOPMENT_CORPUS_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
18	DN12-CXG-06	CELLxGENE	433700dc-e8a5-48b0-b5cd-beb22f3f88fe	VALID_DESIGN	NONE_STRUCTURE_CLEAR	DEVELOPMENT_CORPUS_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
19	DN12-CXG-07	CELLxGENE	17481d16-ee44-49e5-bcf0-28c0780d8c4a	STRUCTURALLY_UNRESOLVED	ESTIMAND.UNRESOLVED	DEVELOPMENT_CORPUS_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
20	DN12-CXG-08	CELLxGENE	e33ffcd3-7cbf-4b8c-b0f4-85587ad5019a	STRUCTURALLY_UNRESOLVED	ESTIMAND.MULTI_AXIS_UNRESOLVED	DEVELOPMENT_CORPUS_EXACT_ID_OVERLAP_FAIL_CLOSED	NO	PREDEFINED_DEVELOPMENT_AND_CALIBRATION_STUDY_FAMILY_OVERLAP	0
21	DN12-AE-01	BioStudies/ArrayExpress	E-MTAB-5905	VALID_DESIGN	NONE_STRUCTURE_CLEAR_PENDING_SDRF_RECOMPUTATION	NO_EXACT_ID_MATCH_IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INCCLUDED	1
22	DN12-AE-02	BioStudies/ArrayExpress	E-MTAB-6653	STRUCTURALLY_UNRESOLVED	SUPPORT.INDEPENDENT_UNITS.BELOW_FROZEN_THRESHOLD	NO_EXACT_ID_MATCH_IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INCCLUDED	1
23	DN12-AE-03	BioStudies/ArrayExpress	E-MTAB-10296	HARD_INVALID_DESIGN	DESIGN.CELL_LINE_METHOD_DEMONSTRATION	NO_EXACT_ID_MATCH_IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INCCLUDED	1
24	DN12-AE-04	BioStudies/ArrayExpress	E-MTAB-5480	HARD_INVALID_DESIGN	DESIGN.TECHNICAL_METHOD_COMPARISON	NO_EXACT_ID_MATCH_IN_FROZEN_DEVELOPMENT_129_AUTHORITY	YES	NOT_APPLICABLE_INCCLUDED	1